



COMPUTER ENGINEERING



ESCAP
Economic and Social Commission
for Asia and the Pacific



ESCAP × KMITL Knowledge Workshop

Rathachai Chawuthai





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Rathachai Chawuthai



- Associate Professor at Computer Engineering, KMITL
- Ph.D in Informatics from SOKENDAI (JP)@2016
 - M.Eng from AIT (TH)@2012, B.Eng from KMITL (TH)@2006
- Working Experience⁽²⁰¹²⁻²⁰¹⁶⁾ :
 - Thomson Reuters (TH)^{4yrs}, Punsarn Asia (TH)^{1yr}, NII (JP)^{3yrs}
- Fields of Interest
 - AI, Machine Learning
 - Large Language Model
 - GPS Data Analytics
 - Software Engineering
 - Semantic Technology

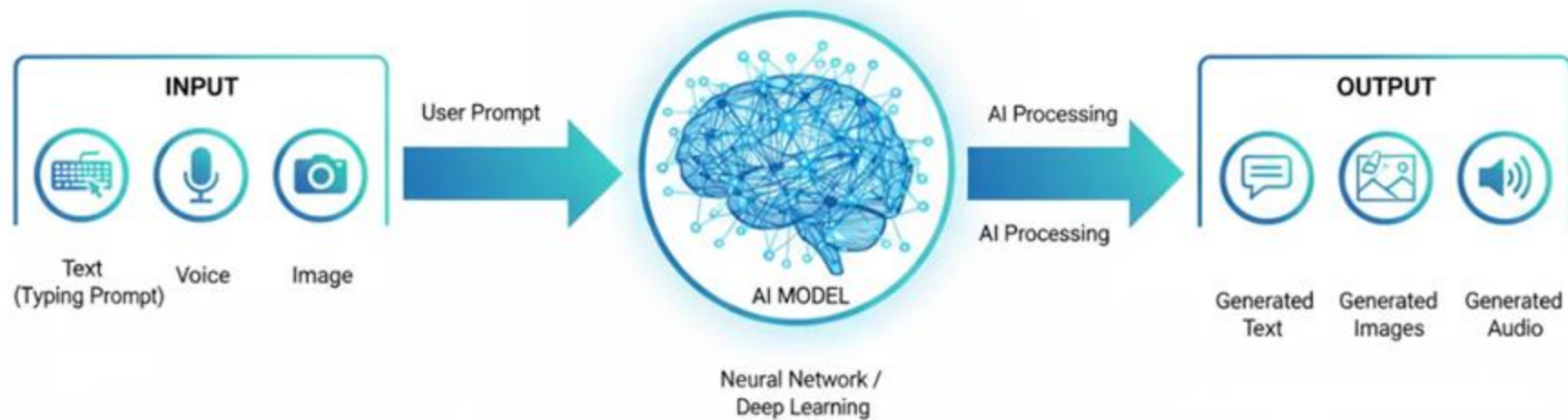


Topics

- Generative AI
- LLMs
- RAG

Generative AI

Generative AI



5 Types of Generative AI



1) Text Generation

CHATGPT, LLM



2) Image Generation

DALL-E, Midjourney



3) Audio/Speech Generation

Text-to-Speech



4) Video Generation

Runway ML, Synthesia



5) Code Generation

Copilot, AlphaCode

LLMs

Large Language Model



My favorite food is a

Input

Output

My favorite food is a

My favorite food is a **bagel**

My favorite food is a bagel **with**

My favorite food is a bagel with **cream**

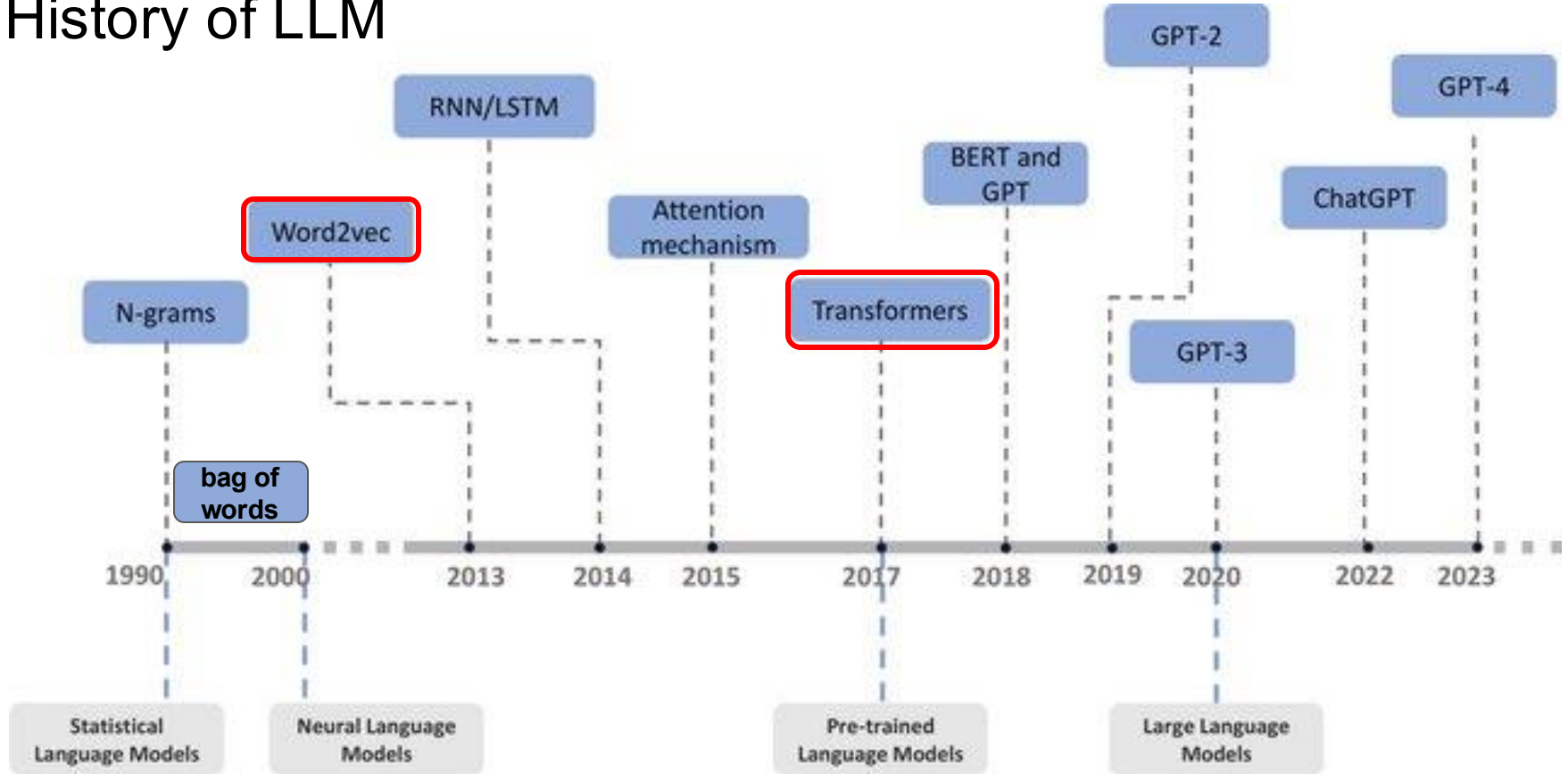
bagel

with

cream

cheese

History of LLM

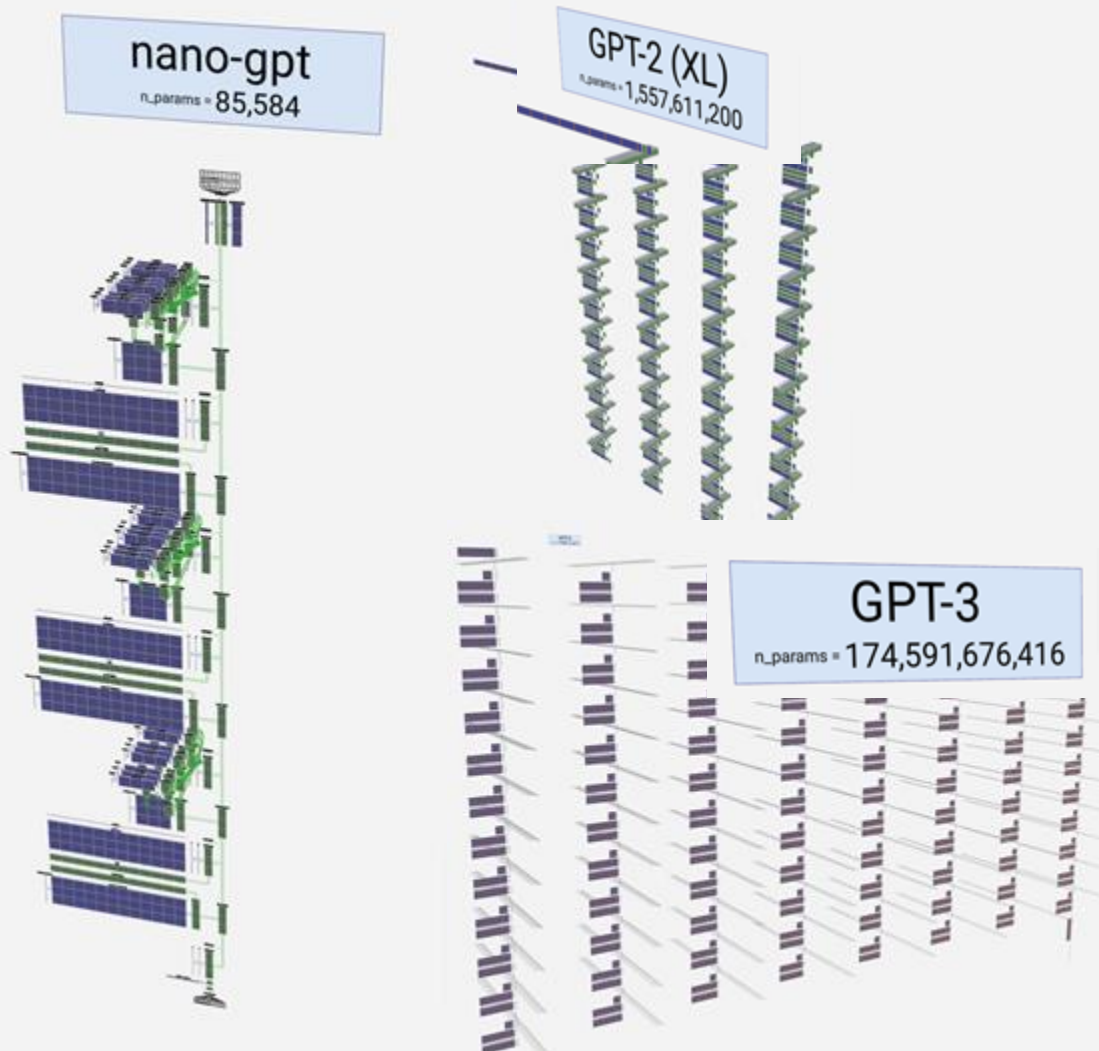


LLM Architecture

<https://bbycroft.net/llm>

<https://www.youtube.com/watch?v=7xTGNNLPyMI>

<https://www.byhand.ai/p/deepseek-lecture-128>



AI Chatbot

Large Language Model

LLM

answer



←

LLM

[illegible]

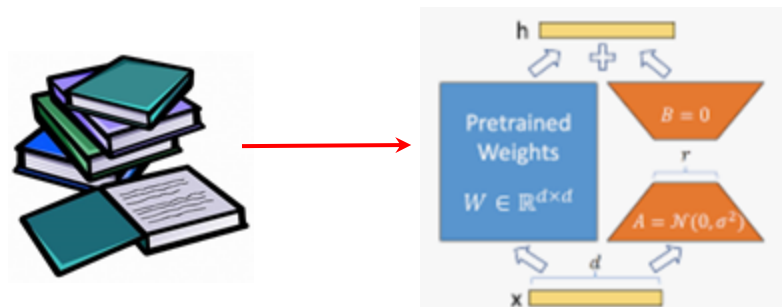
Application

What is CE-KMITL?

llama-3.2-3b-instruct

CE-KMITL, or College of Engineering - King Mongkut's Institute of Technology Ladkrabang, is a college within the university KMITL.

LLM Fine-Tuning

[illegible]

Documents

LLM Fine-Tuning

instruction

Identify the full name of the abbreviation.

input

What is CE-KMITL?

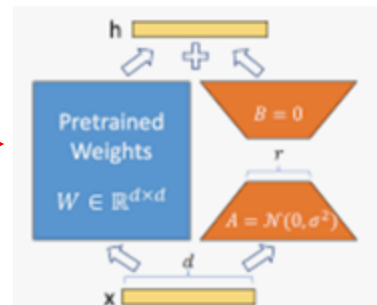
output

The Department of Computer Engineering,
King Mongkut's Institute of Technology
Ladkrabang



Documents

LLM



LLM Fine-Tuning

instruction

Provide the name of an over-the-counter drug commonly used to relieve headaches.

input

I have a mild headache.

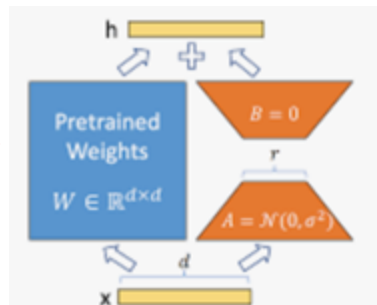
output

You could consider taking Ibuprofen.

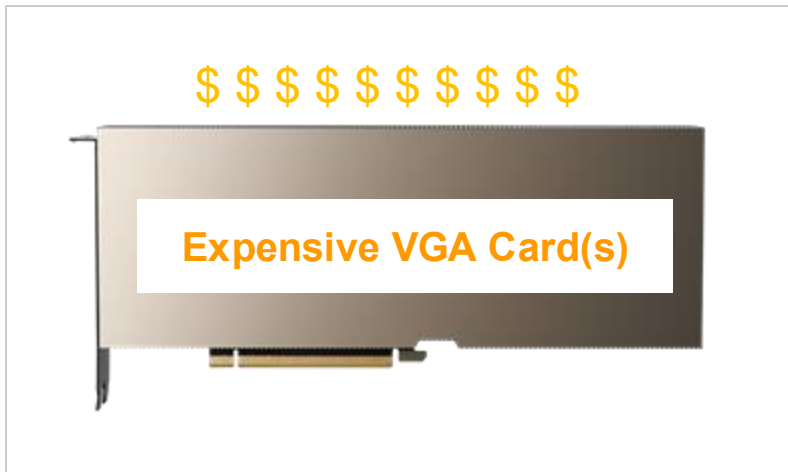
LLM



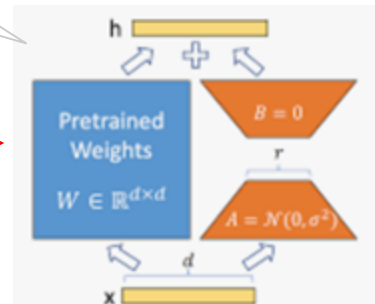
Documents



LLM Fine-Tuning



Documents



RAG

RAG

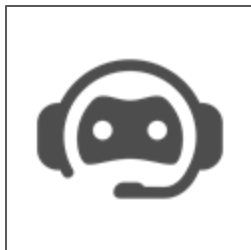
- Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a large language model, so **it references an authoritative knowledge base outside of its training data sources before generating a response.**
- Large Language Models (LLMs) are trained on vast volumes of data and use billions of parameters to generate original output for tasks like answering questions, translating languages, and completing sentences. RAG extends the already powerful capabilities of LLMs to specific domains or an organization's internal knowledge base, all **without the need to retrain the model.**
- It is a cost-effective approach to improving LLM output so it remains relevant, accurate, and useful in various contexts.

RAG

- Known challenges of LLMs include:
 - Presenting false information when it does not have the answer.
 - Presenting out-of-date or generic information when the user expects a specific, current response.
 - Creating a response from non-authoritative sources.
 - Creating inaccurate responses due to terminology confusion, wherein different training sources use the same terminology to talk about different things.

RAG

Chatbot



Application



. Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.



Vector Database

LLM



Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

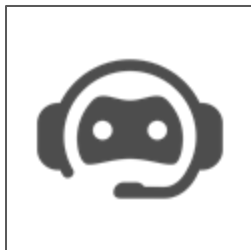
Documents

RAG



question
→

Chatbot



Application

LLM



... Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.



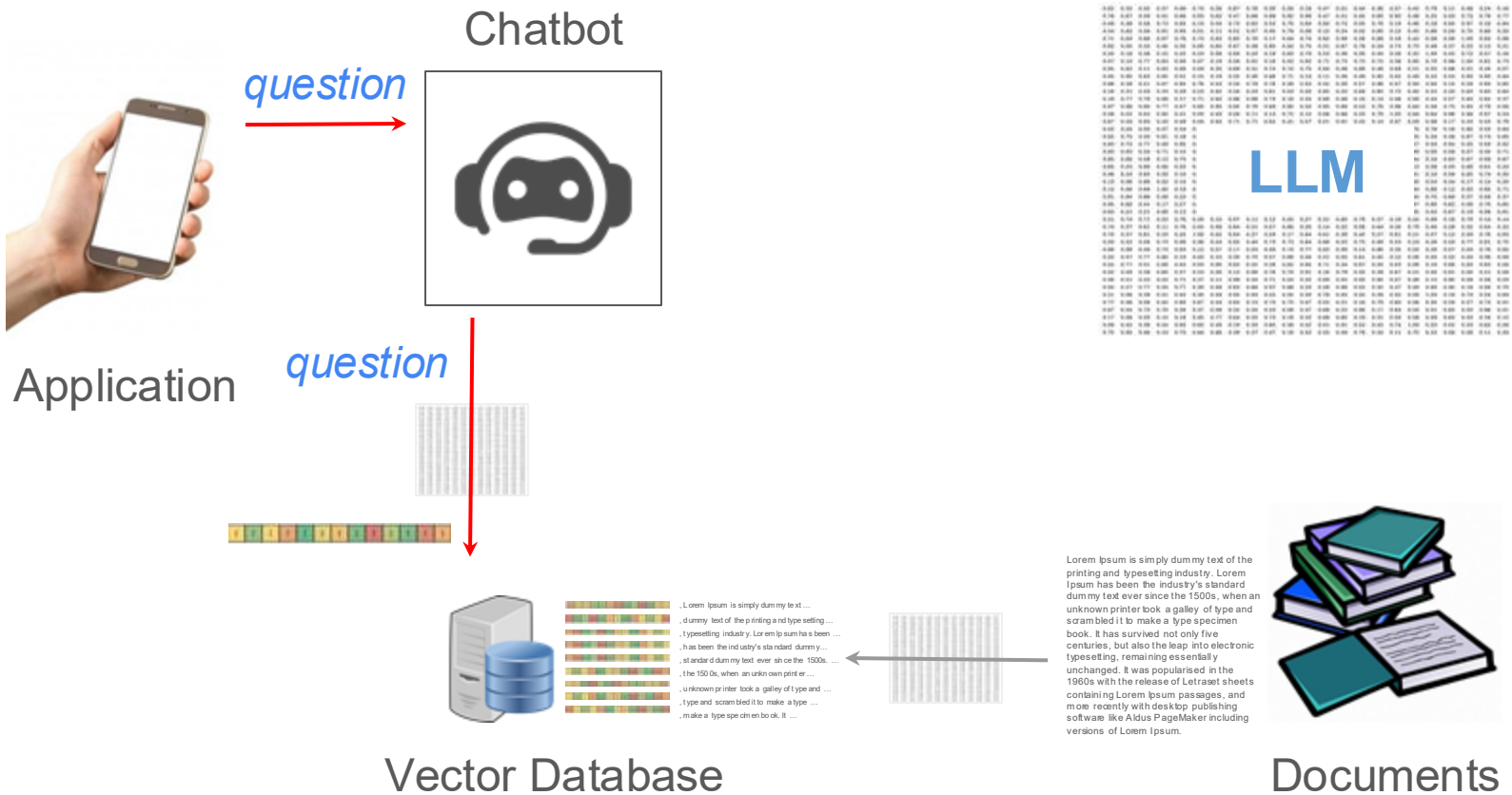
Vector Database



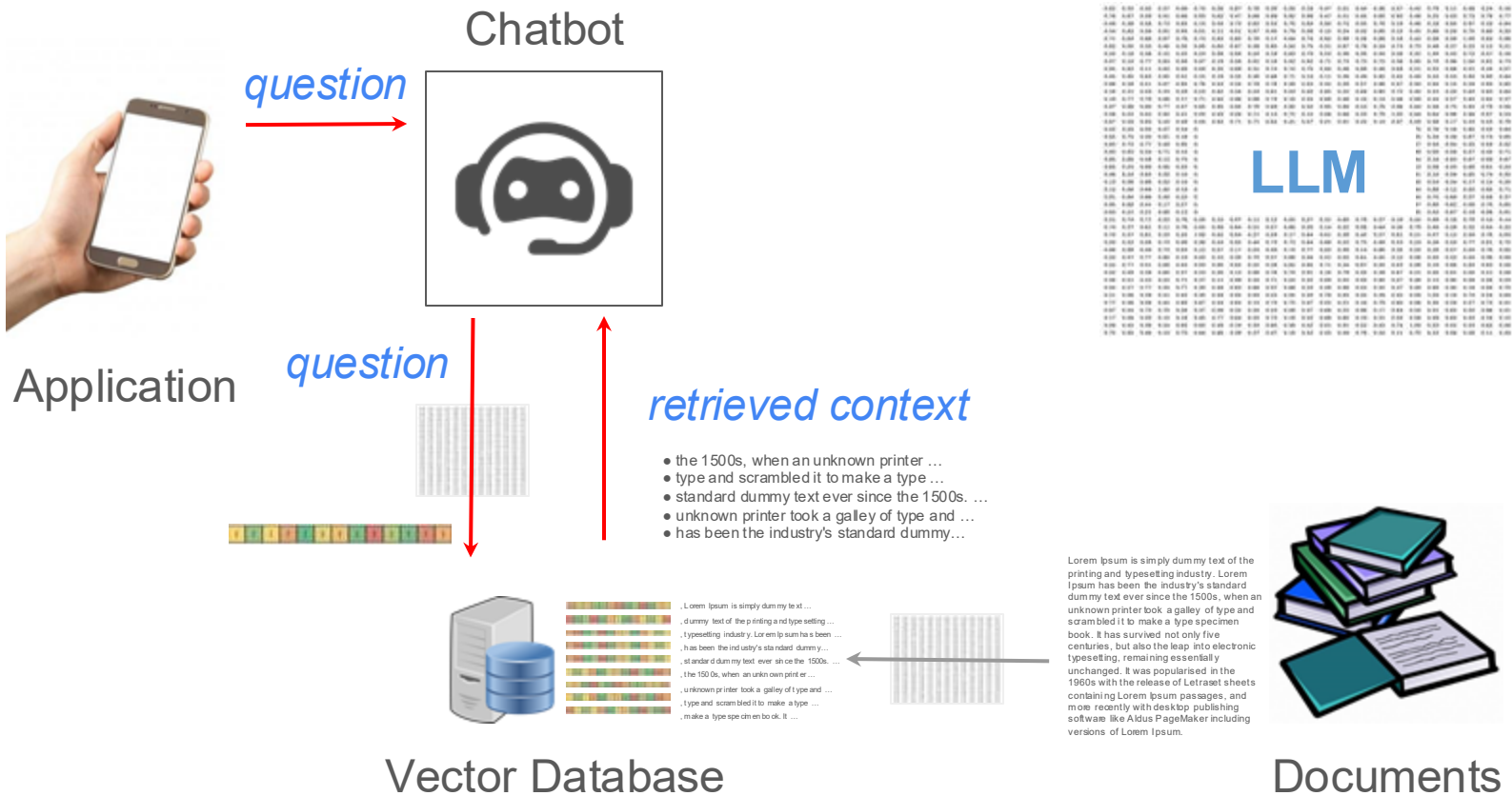
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Documents

RAG



RAG



RAG

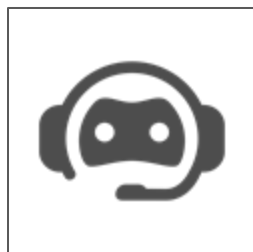


Application

question

question

Chatbot



prompt =
question
+ retrieved context

retrieved context

- the 1500s, when an unknown printer ...
- type and scrambled it to make a type ...
- standard dummy text ever since the 1500s. ...
- unknown printer took a galley of type and ...
- has been the industry's standard dummy...



Vector Database

. Lorem ipsum is simply dummy text ...
. dummy text of the printing and type setting ...
. typesetting industry. Lorem ipsum has been ...
. has been the industry's standard dummy ...
. standard dummy text ever since the 1500s. ...
. the 1500s, when an unknown printer ...
. unknown printer took a galley of type and ...
. type and scrambled it to make a type ...
. make a type specimen by ...



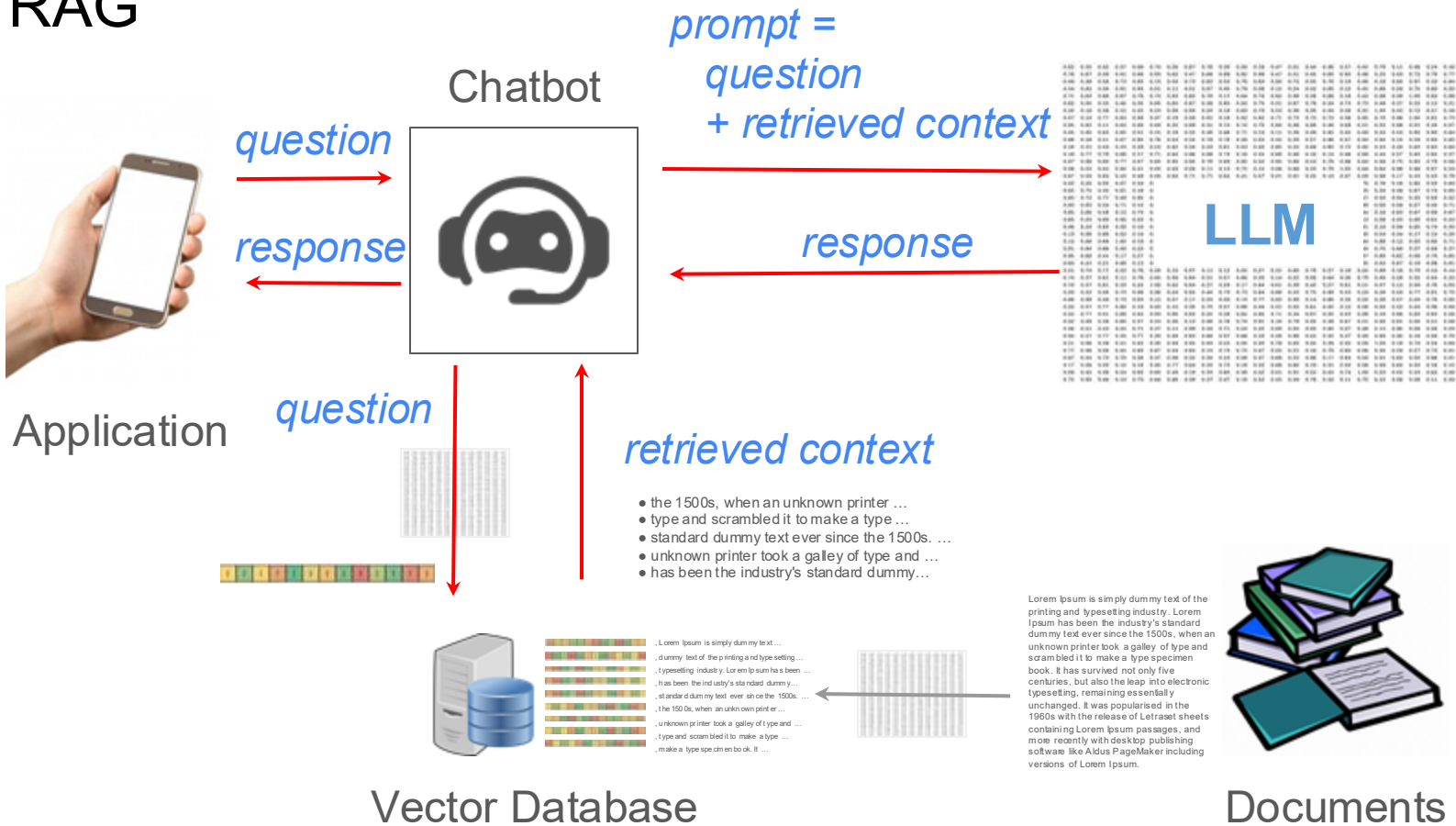
Lorem ipsum is simply dummy text of the printing and typesetting industry. Lorem ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem ipsum.



Documents

LLM

RAG



RAG

4

prompt =
question
+ retrieved context

1

Chatbot

question

response

5

response

LLM

Application

question

2

retrieved context

3

- the 1500s, when an unknown printer ...
- type and scrambled it to make a type ...
- standard dummy text ever since the 1500s. ...
- unknown printer took a galley of type and ...
- has been the industry's standard dummy...

0

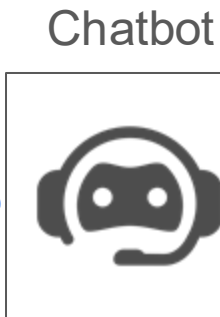
Knowledge Base

Documents

RAG



1
question
response



Chatbot

4

prompt =
question
+ retrieved context

Augmented

5

response

Generation

LLM

Application

question

2

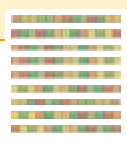


retrieved context

3

- the 1500s, when an unknown printer ...
- type and scrambled it to make a type ...
- standard dummy text ever since the 1500s. ...
- unknown printer took a galley of type and ...
- has been the industry's standard dummy...

Retrieval



Knowledge Base

0

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

To improve the better retrieval process here!



Documents

Example Prompt Template

Use the following context to answer the question below.

Context:

{{Retrieved Context}}

Question: **{{User Query}}**

Answer:

You are an expert in **{{application domain}}**.

Please answer the following question in **{{language}}**. (Giving answer only)

Use only the provided context to answer the question below.

Context:

{{Retrieved Context}}

Question: **{{User Query}}**

Example Prompt for RAG

- Copy this prompt into any LLM

You are an expert in **Food**.

Please answer the following question in **English**. (Giving answer only)

Use only the provided context to answer the question below.

Context:

- **Basic okonomiyaki batter: flour, dashi, egg, nagaimo. Mix well.**
- **Key fillings: shredded cabbage, thinly sliced pork belly, seafood. Combine with batter**
- **Cook on a hot griddle. Pour batter, form round shape. Brown one side.**
- **Flip carefully. Cook until internal ingredients are done and both sides are golden.**
- **Top with okonomiyaki sauce, mayo, bonito flakes, and nori for serving.**

Question: **When can I finish cooking Japanese Pizza?**

Example Prompt for RAG

- Copy this prompt into any LLM

You are an expert in **KMITL**.

Please answer the following question in **English**. (Giving answer only)

Use only the provided context to answer the question below.

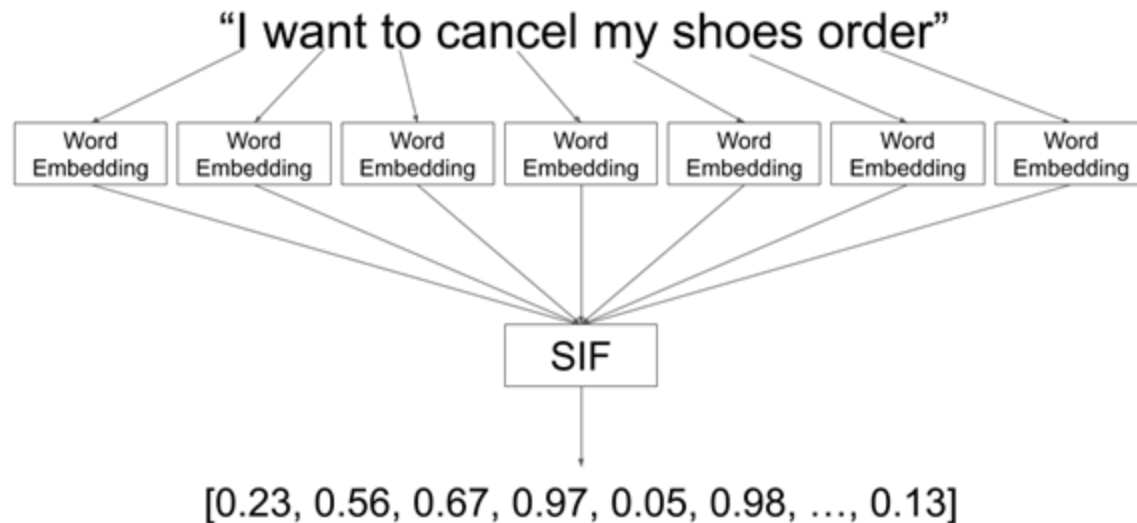
Context:

- is held as a memorial to Chaophraya Surawong Waiyawat, commonly known as "Chao Kun Thahan", as Ms. Liam Protpittayapayat
- as "Chao Kun Thahan", as Ms. Liam Protpittayapayat, his heir, stated her intentions in donating
- on which the institute is currently located, with 1,345.6 square kilometers used from
- 1,345.6 square kilometers used from a total of 1,665.6 square kilometers donated
- donated (the remaining 320 square kilometers was donated to Protpittayapayat

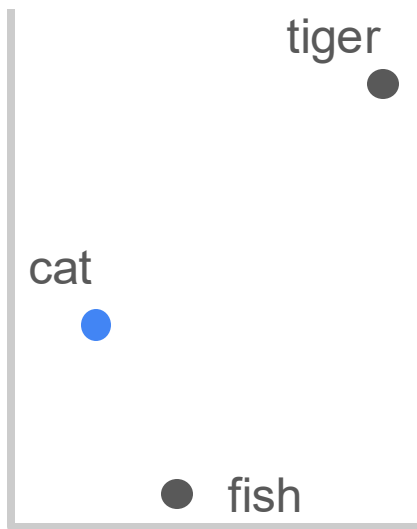
Question: Who donated the land to KMITL?

Vector Database

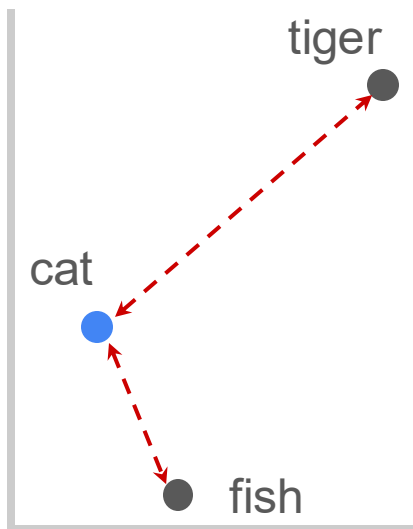
Sentence Embedding



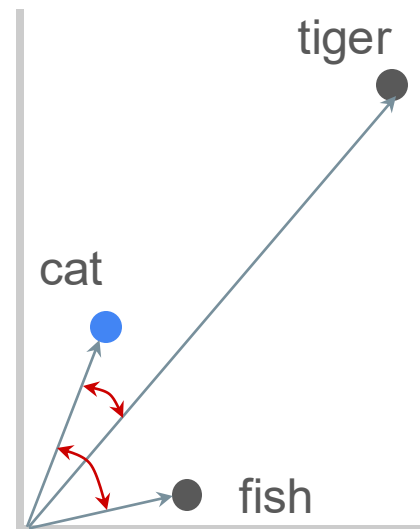
Distance Matrices



Token Vectors

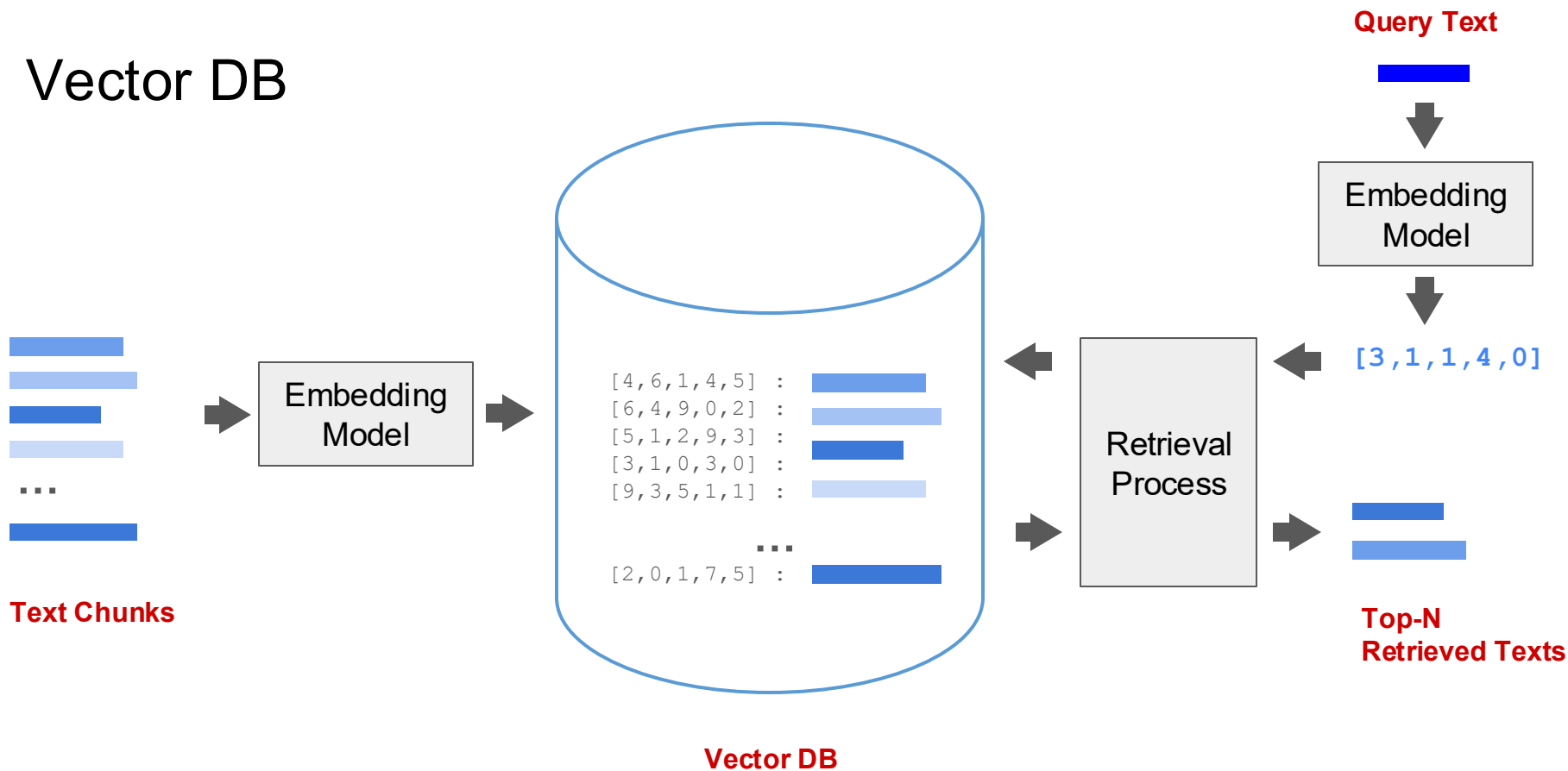


Euclidean Distance



Cosine Distance

Vector DB



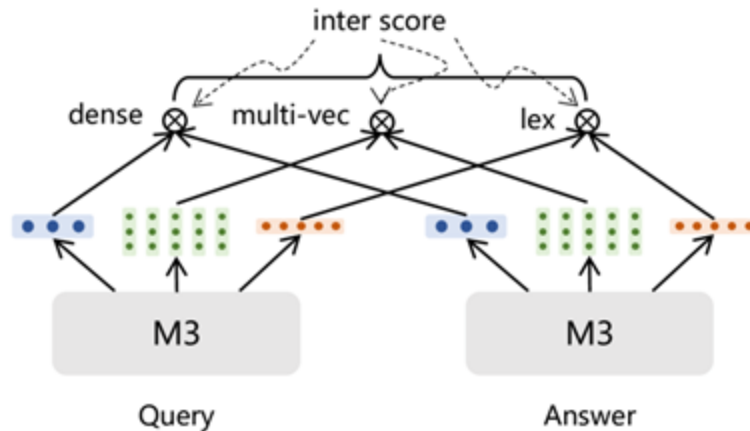
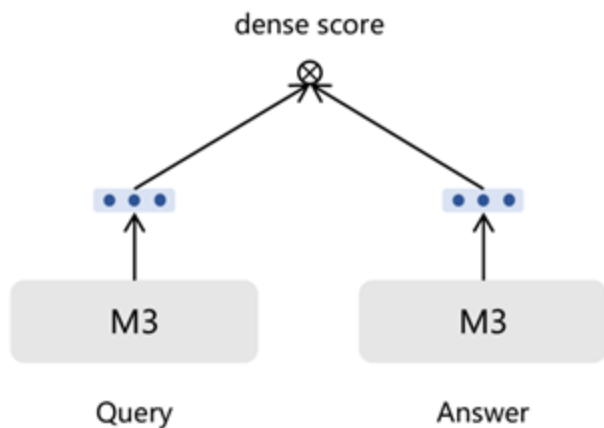
BGE-M3 : Embedding Model

BGE-M3 is a compound and powerful embedding model distinguished for its versatility in:

- Multi-Functionality: It can simultaneously perform the three common retrieval functionalities of embedding model: dense retrieval, multi-vector retrieval, and sparse retrieval.
- Multi-Linguality: It can support more than 100 working languages.
- Multi-Granularity: It is able to process inputs of different granularities, spanning from short sentences to long documents of up to 8192 tokens.

Try: <https://huggingface.co/BAAI/bge-m3>

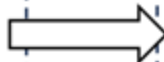
BGE-M3 : Embedding Model



Pre-Training



1.2 Billion multilingual
unsupervised data



Fine-Tuning



- Labeled data (EN, ZH, Mul)
- Synthetic data (Long)

[Hand-On] Vector Search & RAG



Vector Search & RAG (English Version)

- <https://colab.research.google.com/drive/1NI68S7VZbz2kFfoGWZ7E8dKXdi-zros2>

```
def context_search(query):  
    qvec = get_embedding(query)  
    print('query vector = ', qvec.round(3))  
    vecdb['sim'] = vecdb['vec'].apply(lambda x: cosine_similarity(x, qvec) )  
    return vecdb.sort_values('sim', ascending=False)[['sim', 'text', 'vec']]
```

```
context_search('Can student submit befor half hour?')
```

```
query vector = [-0.014 -0.014  0.024 ... -0.031 -0.049  0.001]
```

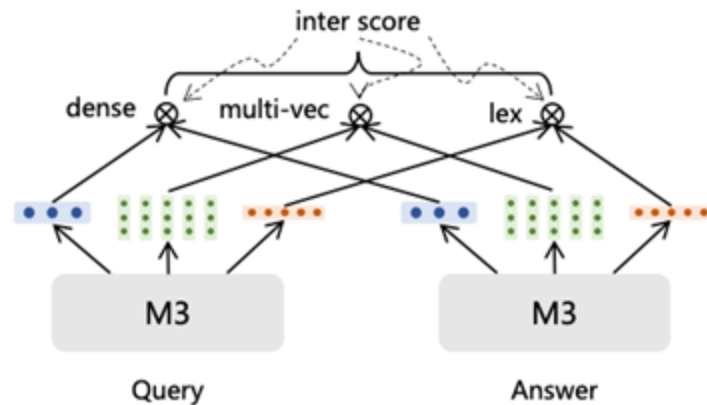
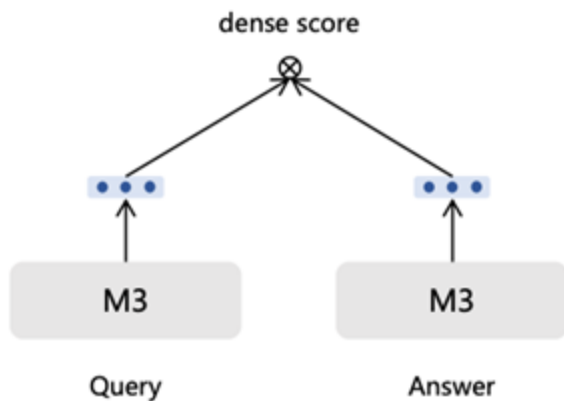
sim

text

7	0.777284	No student is allowed to leave the exam room during the first thirty minutes or the final fifteen minutes of the testing period.	[0.018326333, -0.00937139 -0.021894662, 0.02055
8	0.749968	If a student requires a fresh piece of scrap paper or a replacement pen, they must raise their hand and wait for an invigilator to assist them.	[-0.0014063054, -0.0036780 -0.018697433, 0.028569
4	0.749013	Students must write their full name and student identification number clearly on the cover page of the answer booklet.	[-0.007813321, -0.0102745 0.024256986, -0.017

More Things

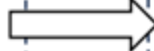
BPE-M3



Pre-Training



1.2 Billion multilingual
unsupervised data

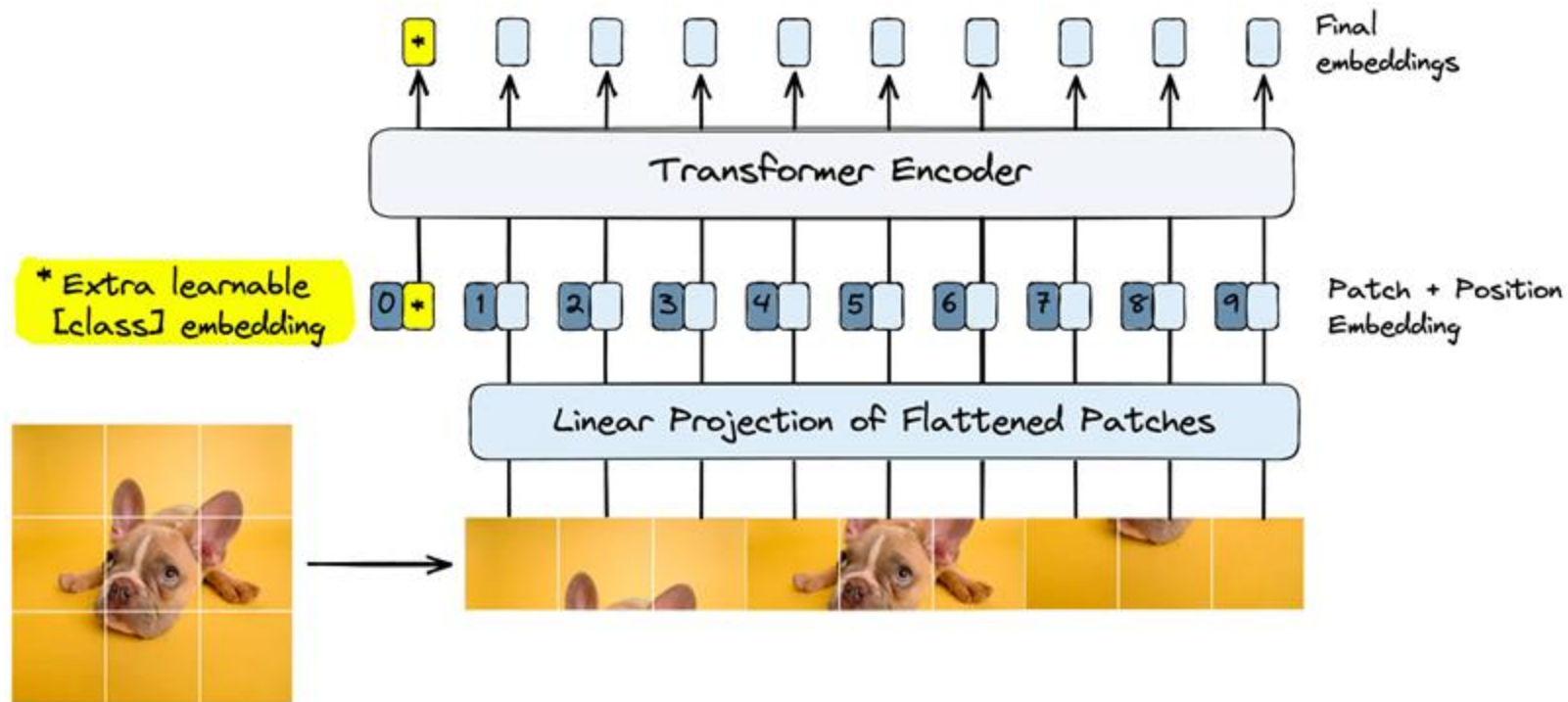


Fine-Tuning

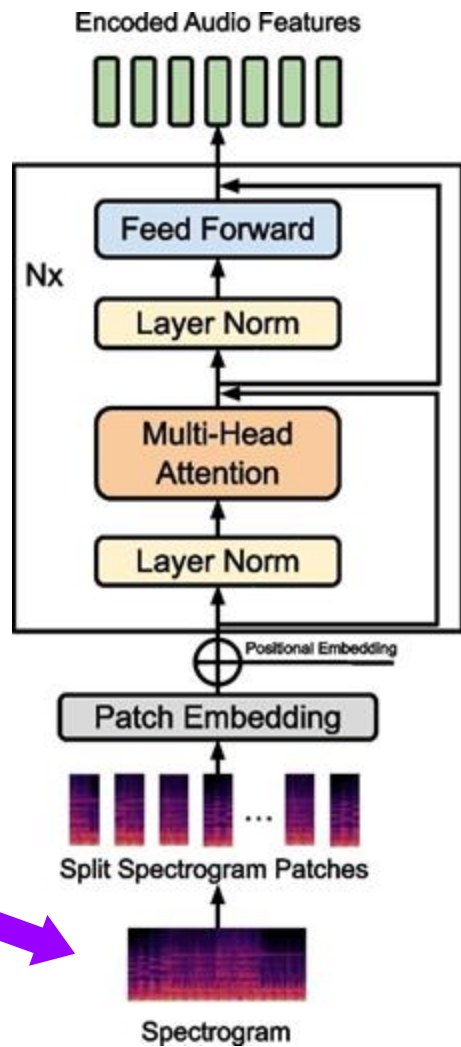
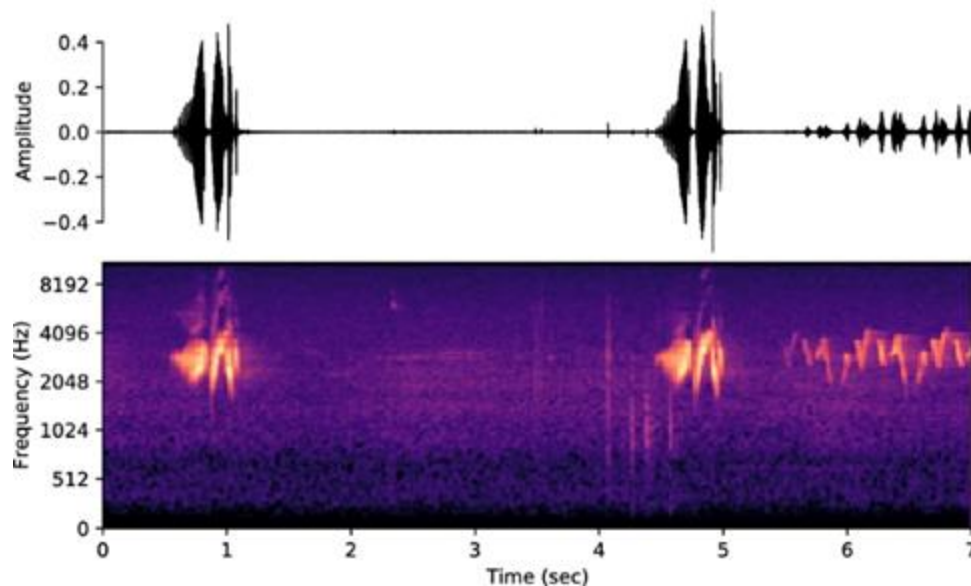


- Labeled data (EN, ZH, Mul)
- Synthetic data (Long)

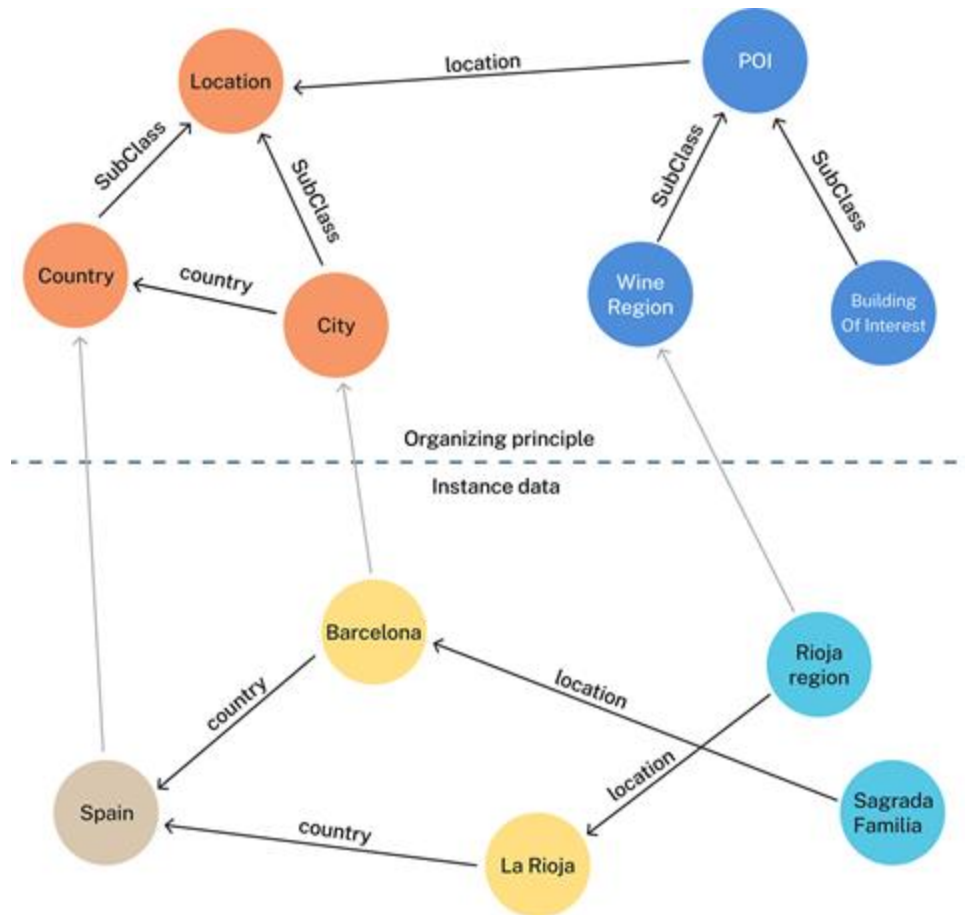
Vision Transformers (ViT)



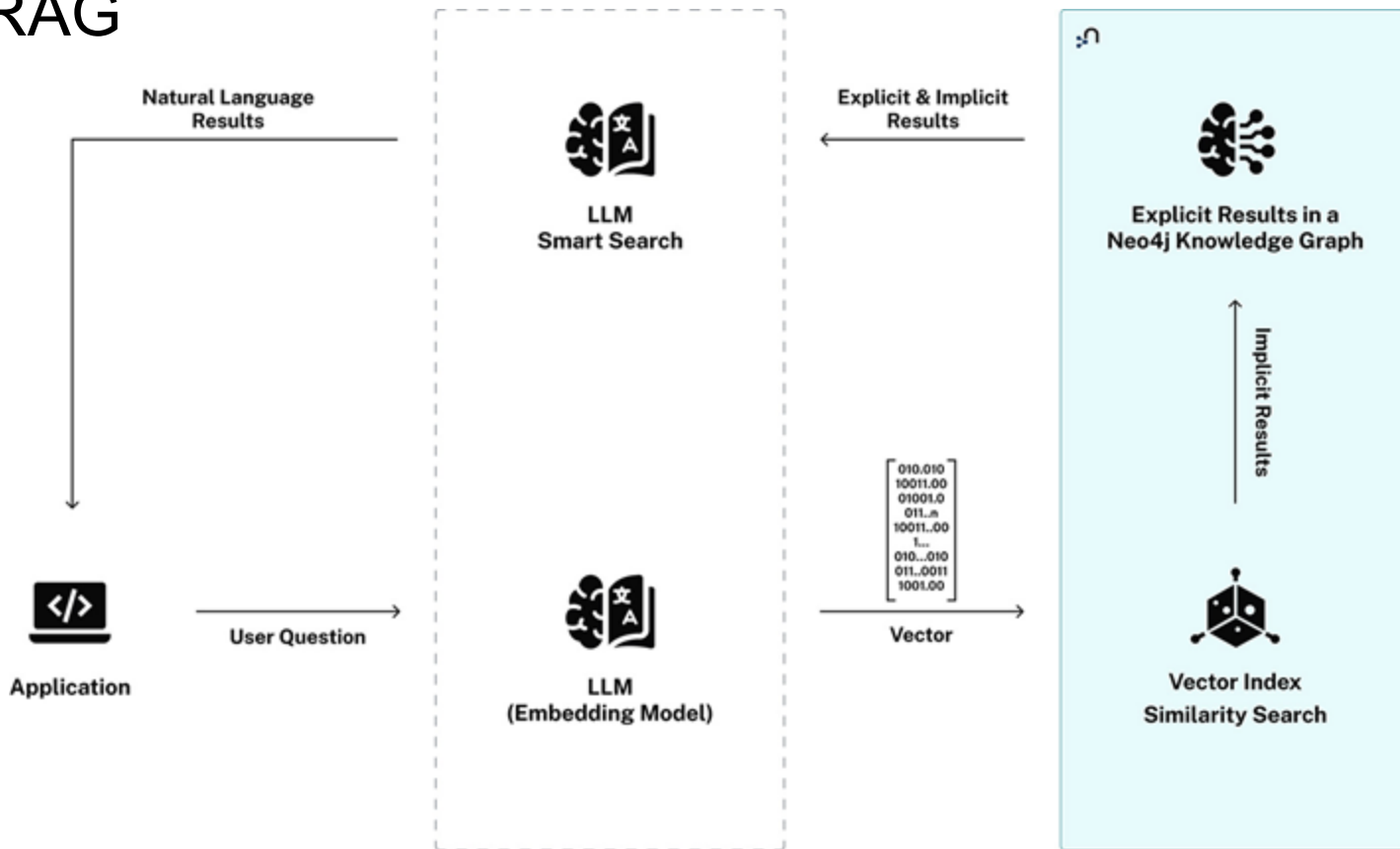
Audio Embedding



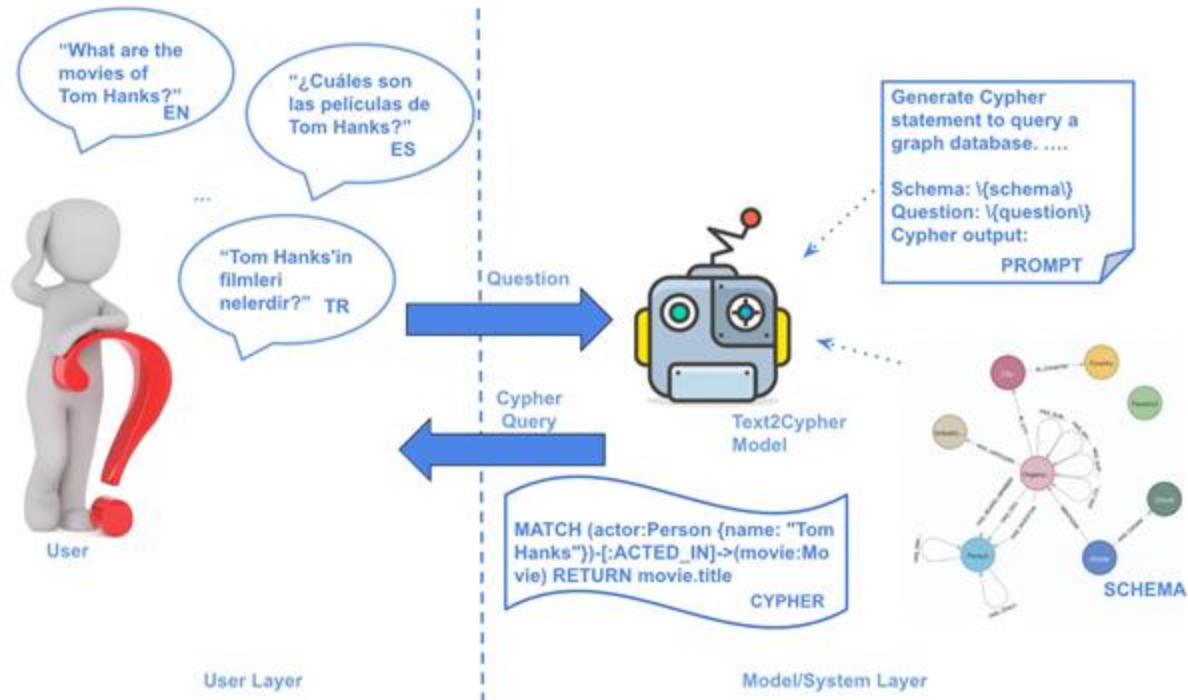
Knowledge Graph



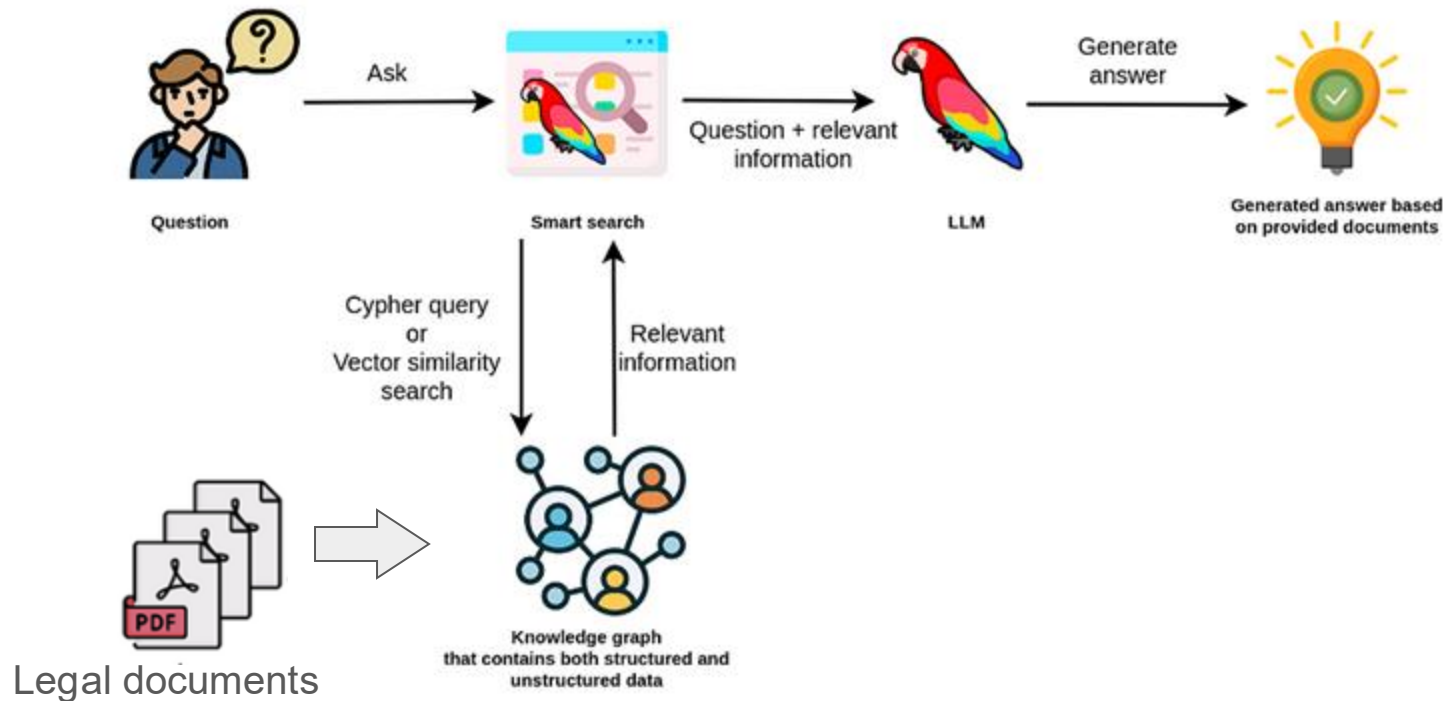
Graph RAG



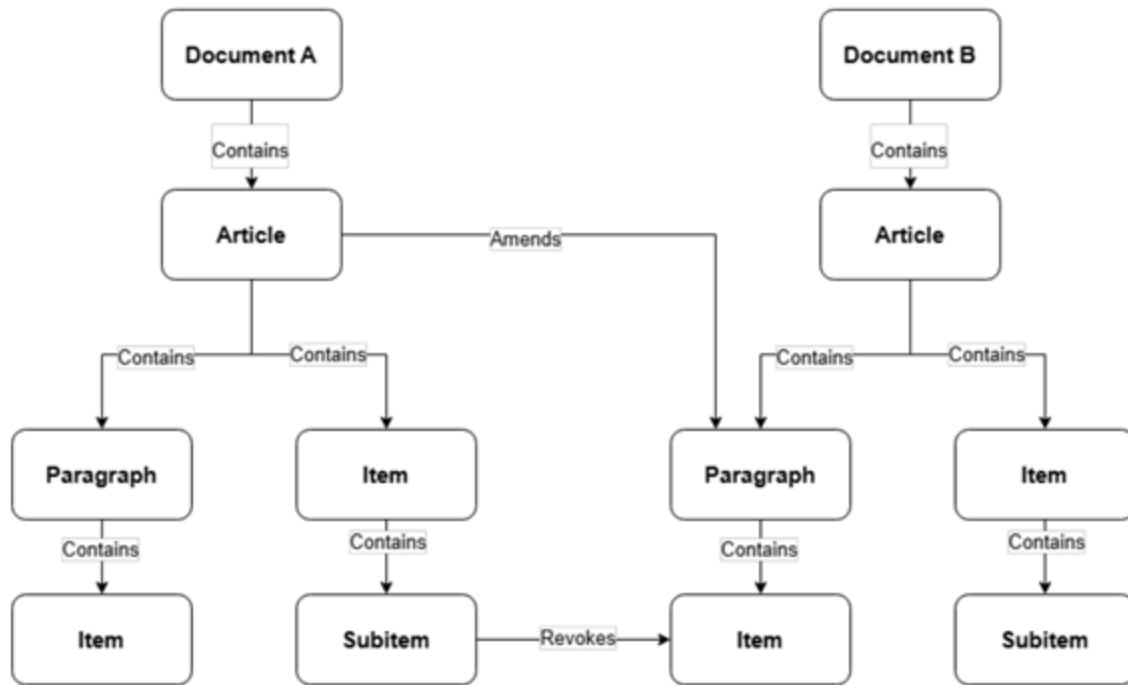
Text-to-Cypher (neo4j)



GraphRAG

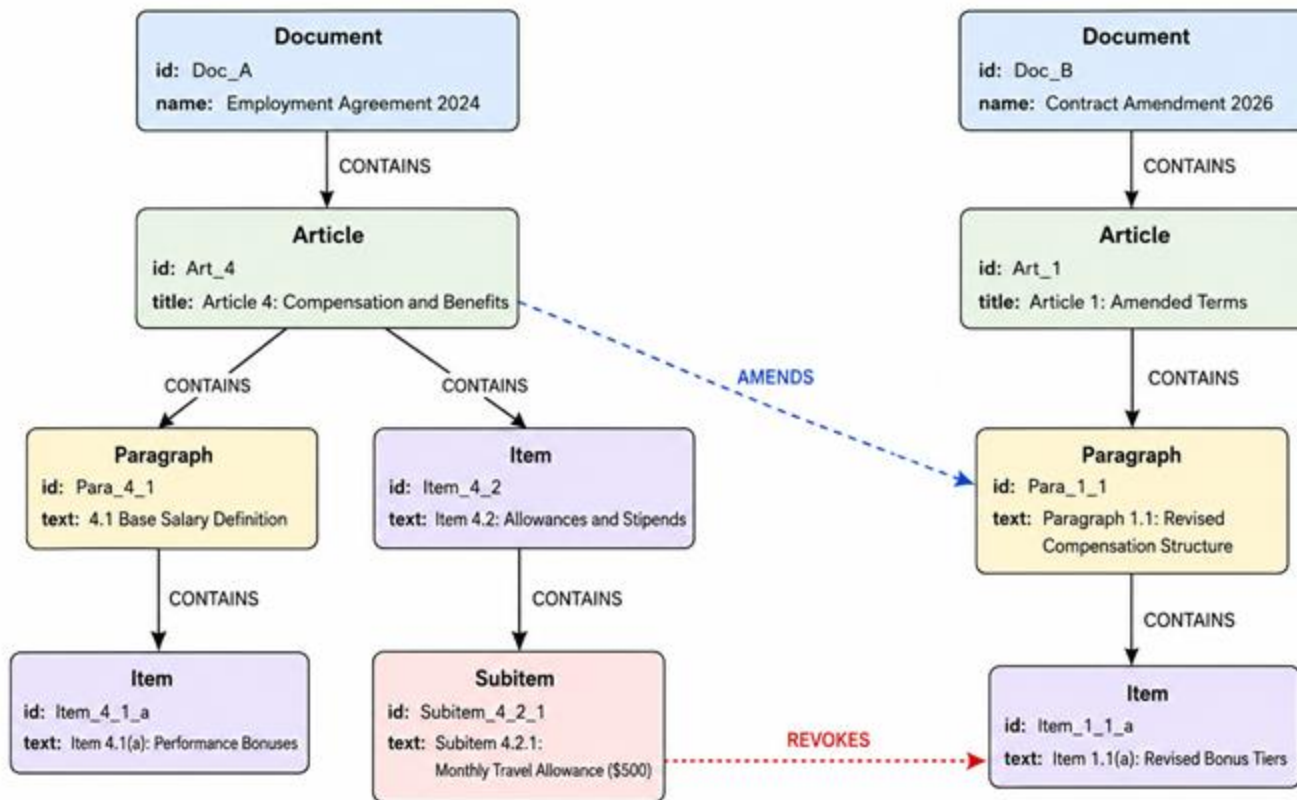


GraphRAG for Legal



Ref: de Oliveira, V. T., da Silva, D. O., de Almeida Souza, M., Lima, M. R., de Oliveira, S. S. T., & Rosa, T. C. (2026, April). Retrieval-Augmented Generation and Knowledge Graphs in Portuguese-Language Legal Documents. In Proceedings of the 17th International Conference on Computational Processing of Portuguese (PROPOR 2026)-Vol. 1 (pp. 1-10).

GraphRAG for Legal



Microsoft GraphRAG

The screenshot displays the LM Studio application window. At the top, it shows 'Model RAM Usage: 12.01 GB' and 'CPU: 0.00 %'. The main interface is divided into several panels:

- Local Inference Server:** This panel on the left provides instructions on starting a local HTTP server. It lists supported endpoints: `/v1/chat/completions`, `/v1/embeddings`, and `/v1/models`. It also includes a link to 'API Documentation: Local Server API / C-opts in browser'. The 'Configuration' section allows setting the 'Server Port' to 1234, enabling 'Cross-Origin Resource Sharing (CORS)', 'Verbose Server Logs', and 'Apply Prompt Formatting'. 'Start Server' and 'Stop Server' buttons are at the bottom.
- Embedding Model Settings:** This central panel shows the 'Embedding Model Settings'. It instructs to 'Load a test embedding model and utilize it through the `POST /v1/embeddings` endpoint.' The selected model is 'CompendiumLabs • bge-large-en-v1.5-f32-335M-F32-gguf'. There is a checkbox for 'Offload entire model to VRAM' which is checked. Below this, there are 'Examples' tabs for different use cases, with 'embeddings (python)' selected. A code block shows a Python script for using the OpenAI client to get embeddings. A 'Copy Code' button is at the bottom right of the code block.
- Server Model Settings:** This panel on the right shows the 'Server Model Settings'. It includes a 'Presets' section with 'Llama 3 V2' selected. There are buttons for 'Danger Zone', 'Discard Changes', and 'Override Preset'. The 'System Prompt' section contains a default prompt: 'You are a helpful, smart, kind, and efficient AI assistant. You always fulfill the user's requests to the best of your ability.' There is also an 'Ask before overriding' checkbox. The 'Advanced Configuration' section includes settings for 'Context Length' (set to 131072), 'Temperature' (set to 0.5), 'Tokens to generate' (set to 1), and 'CPU Threads' (set to 4).

At the bottom, the 'Server logs' panel shows the server running on port 1234. It includes a 'Filter logs...' button and 'Open Logs' and 'Clear (Ctrl+R)' buttons. The logs show the server starting, supported endpoints, and the model being loaded.

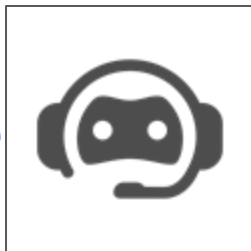
Summary

RAG



1
question
response

Chatbot



4
prompt =
question
+ retrieved context

5
response

Augmented

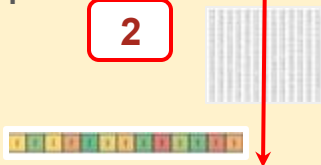
LLM

Generation

Application

question

2

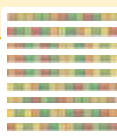


retrieved context

3

- the 1500s, when an unknown printer ...
- type and scrambled it to make a type ...
- standard dummy text ever since the 1500s. ...
- unknown printer took a galley of type and ...
- has been the industry's standard dummy...

Retrieval



Knowledge Base

0

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

To improve the better retrieval process here!



Documents

つづく